

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Digital Gift Card & Loyalty Point Exchange

SRN: PES1UG24CS091 | Section: B | Domain: Retail, E-Commerce & Finance

Users: Account Holder, Merchant Partner

Architecture Selection

We chose Layered Architecture for the Digital Gift Card & Loyalty Point Exchange System.

Two Reasons

1. Clear separation of responsibilities: The architecture separates the presentation, business and data concerns. The User Interface handles Account Holder and Merchant Partner interaction, while the Exchange Manager coordinates the core exchange workflow.

2. Centralized exchange rules: The Exchange Manager can coordinate the Jira-defined workflow of validating loyalty points, deducting points, converting points to exchange credits using a dynamic conversion rate, checking eligible digital gift cards, and coordinating redemption. This keeps the business rules consistent across exchanges.

Security Advantage

The business layer provides a controlled boundary between users and stored transaction data. Exchange requests can be validated before points, credits, gift-card eligibility and voucher records are changed. This supports the security-related work represented in Jira, including voucher validation and preventing duplicate redemption.

Performance Benefit

The architecture keeps the main exchange flow direct: the User Interface communicates with the Exchange Manager, which coordinates the required service and database operations. Avoiding the additional network and operational overhead of a distributed microservices design makes the architecture suitable for responsive exchange processing and supports the system's strict performance objective.

Selected components: User Interface, Exchange Manager, Loyalty & Exchange Service, Merchant Partner Service, Database.

Required interfaces: Exchange Request, Loyalty & Exchange, Merchant Transaction, Data Persistence.